Abinaya G
Department of Information Technology
Saveetha Engineering College
Chennai, India

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The Editors-in-Chief
Expert Systems with Applications

Dear Editors,

I submit the manuscript *"EIG-MPHBS: a rotation-invariant information exchange for hybrid Honey Badger–Marine Predators optimisation"* for consideration as a research article in *Expert Systems with Applications*.

Hybrid metaheuristics are widely used as the optimisation engine of intelligent systems in engineering. Many of them let two populations cooperate by recombining their members coordinate by coordinate, an exchange that breaks down when the variables of a problem are coupled. The manuscript proposes EIG-MPHBS, a hybrid Honey Badger–Marine Predators optimiser whose information exchange is carried out in the eigenvector frame of the pooled populations.

The main contributions are:

- EIG-MPHBS, with a proof that its exchange is invariant to rotations of the search space and uses no extra function evaluations;
- a mechanism analysis showing that the eigen frame keeps the exchange effective on coupled problems, where coordinate-wise exchange stalls;
- an evaluation in 18 840 seed-paired runs on CEC2022, the 24 BBOB functions and eight FIR filter-design problems against ten algorithms, including the parent algorithms, PSO, DE, GWO, WOA, L-SHADE and CMA-ES;
- a design fixed before the experiments, with confirmatory tests on held-out instances, and public code, run records and analysis scripts.

EIG-MPHBS is never significantly worse than the original hybrid on any of the 56 instances, reduces the error on the rotated ellipsoid fourteen-fold, and outperforms HBA, MPA, PSO, DE, GWO and WOA in 260 of 336 comparisons, at no extra evaluation cost. The FIR filter-design problems demonstrate its use in an engineering application.

The manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. The author declares no competing interests, and the research received no specific funding.

Thank you for considering this submission.

Yours sincerely,

Abinaya G
Department of Information Technology
Saveetha Engineering College,
Chennai, India
gabinaya@saveetha.ac.in